

Unified Theory of Adaptive Meaning — Part II

Will as a Formal Operator in the NC2.5 Axiomatic Core

Author: Maksim Barziankou (MxBv)

Contact: research@petronus.eu

Date: 2026-04-18

License: CC BY-NC-ND 4.0

DOI: [10.5281/zenodo.19646174](https://doi.org/10.5281/zenodo.19646174)

Axiomatic Core (NC2.5 v2.1): [10.17605/OSF.IO/NHTC5](https://doi.org/10.17605/OSF.IO/NHTC5)

Attribution: petronus.eu

Will as a Formal Operator in the NC2.5 Axiomatic Core

PETRONUS™ Research Series, 2026

DOI: [\[10.5281/zenodo.19646174\]](https://doi.org/10.5281/zenodo.19646174)(<https://doi.org/10.5281/zenodo.19646174>)

> *"The chaos around us is merely a limitation of our cognitive ability to hold the integrity of the picture across time."*

>

> — A sentence written during a walk in a forest in Poznań, before any of this had a name.

Document Role

This is a **bridge document**. It is not a proof paper, not a textbook chapter, and not a standalone formalization. Its purpose is to show that three bodies of work — the philosophical (UTAM + ONTOΣ series), the experiential (*Through a Life*), and the mathematical (NC2.5 axiomatic core, v2.1, 61 axioms, 69 theorems, 21 lemmas) — describe the same architecture in three different languages.

The formal proofs referenced in this document are developed in full in the NC2.5 axiomatic core (DOI: [10.17605/OSF.IO/NHTC5](https://doi.org/10.17605/OSF.IO/NHTC5)). The engineering instantiations are developed in the patent portfolio and the bridge papers published on petronus.eu. This document does not repeat those proofs. It reveals the structural identity between them.

Everything began with UTAM. UTAM Part I (2025) was the first attempt to name what was seen in a forest in Poznań: that adaptive systems navigate within a geometry of meaning, and that departures from that geometry are irreversible. The ONTOΣ series, the engineering works, the patents, and the axiomatic core all grew from that seed. UTAM Part II returns to the origin and shows that the seed contained the entire tree.

The UTAM series will continue to lead the expansion of the Unified Theory of Adaptive Meaning. Each new extension of NC2.5 — every new theorem, every new bridge result, every new engineering instantiation — will be reflected back through UTAM as the unifying narrative layer of the theory.

First work in the series: [Unified Theory of Adaptive Meaning — Part I](<https://petronus.eu/blog/unified-theory-of-adaptive-meaning-utam-will-coherence-and-drift-as-the/>)

Structural note: The UTAM series, the ONTOΣ series, and the NC2.5 axiomatic core (current version: v2.1) are best understood together. UTAM provides the narrative and meaning-layer. ONTOΣ provides

the ontological depth. NC2.5 provides the formal apparatus — axioms, theorems, proofs. Each can be read independently, but the full architecture becomes visible only when all three are in view.

Abstract

Part I of the Unified Theory of Adaptive Meaning introduced Will as a neutral ontological operator and UTAM as the geometry of meaning-preserving trajectories in adaptive systems. It drew from Schopenhauer, Bergson, Nietzsche, and Heidegger — but stripped Will of its moral, psychological, and teleological baggage and repositioned it as a structural primitive: the thing that selects admissible directions of becoming.

What Part I could not do was prove that this was more than philosophy.

Part II makes that correspondence explicit. It shows that the meaning-preserving manifold described in Part I is formally identical to the admissible continuation space defined in Navigational Cybernetics 2.5. That the budget of meaning is $\tau = C - \Phi$. That spin — the non-potential component of directed change — is the necessary mechanism by which Will navigates under irreversible structural budget. That the privacy of meaning is non-reconstructibility. That the limit of self-correction is meta-revision under Lyapunov descent.

This is not a second theory built on top of the first. It is a formal correspondence showing that philosophy, lived experience, and mathematics were describing the same architecture from three different entry points.

One theory. Three languages. One axiomatic core.

Section 0 — Before the Formalism

I wrote that sentence while walking my dog through a forest in Poznań. It was not an intellectual exercise. It was a sudden, involuntary recognition — that the disconnected pieces of the world I had been seeing my entire life were not disconnected. They were one structure. I simply could not hold the whole picture at once.

Months later, I learned about the Sierpiński triangle, the Mandelbrot set, and other abstractions that show infinite complexity arising from finite rules. I recognized what I had seen in the forest: not chaos, but coherence beyond my cognitive bandwidth.

What followed was not planned. I began writing.

The earliest works were not systematic. They were recognitions. *Synthetic Conscience: The Emergence of Engineered Vitality Systems* was the compass needle — a paper that now looks like it was pointing toward everything that came after. It asked whether an adaptive system could be given something functionally equivalent to conscience: not a moral module, not a rule set, but an architectural mechanism that translates structural awareness into engineering behavior. It introduced EVS — Engineered Vitality Systems — the idea that vitality is not a biological metaphor but an architectural property that can be designed. And it drew a line that the rest of the corpus would deepen: the distinction between living and non-living systems is not substrate-dependent. It is operator-dependent. A system is "alive" — in the structural sense — if it contains an operator that initiates the cycle of impulse, interpretation, and coherence. A system without that operator is reactive. It responds, but it does not navigate. EVS was the first attempt to say: this operator can be engineered. That paper seems almost lost in the corpus now, but it is the foundation of where the entire program is heading.

Then came *The Synthetic Conscience Effect: How ΔE Translates Awareness into Engineering*. At the time, I did not fully understand what ΔE was. I thought it was a hyperreactive engine based on dynamic coherence — a fast controller that stabilized behavior in chaotic environments. It was only later, as the axiomatic core took shape, that I recognized ΔE for what it actually is: the dissipative mechanism of Will.

Not a controller in the classical sense, but the architectural component that absorbs the rotational excess generated by spin, slowing the rate of structural burden accumulation and extending the system's structural lifetime. The paper showed that awareness of one's own coherence can be made operational — and that was the real breakthrough, even if the full formalism would come much later.

These works were the soil. What grew from them was sharper.

First came the IIC Law — Impulse → Interpretation → Coherence — a recognition that every adaptive system, from a neuron to an organization, passes through the same three-phase cycle: something arrives, the system processes it through its internal structure, and either coherence is restored or something irreversible is recorded. It was not yet formalized. It was an observation that refused to stay an observation — because it kept appearing everywhere, in every system I examined, regardless of substrate.

Then came *Structural Drift as a Fundamental Law of Adaptive Behavior* — the proof that drift is not a failure mode but the structural default of any adaptive system. Drift does not require error. It requires only time and coupling. This was the moment the theory stopped being about control and became about survival.

Then came UTAM — the Unified Theory of Adaptive Meaning — which tried to name what I had seen in the forest and what IIC and Drift had revealed in behavior: Will as a geometry of meaning-preserving trajectories. If a system acts within that geometry, its meaning holds. If it steps outside, something irreversible is lost.

Then came the ontological series — ONTOΣ I through IX — each attempting to deepen the same insight from a different angle: that the world is held together by a directed force that is not energy, not information, not purpose, but something prior to all of them. I called it Will, and I meant it not as a psychological property but as an ontological operator — the thing that selects which directions of becoming are admissible and which are not.

Then came the engineering and bridge works — each extending the theory into territory where philosophy alone could not reach. *Structural Pressure: The Missing Primitive* proved that even non-acting systems consume viability — that merely existing under load is itself a monotone cost, validated against lithium-ion battery calendar aging data. *Transaction vs Structural Admissibility* split the industry's conflation of two fundamentally different objects: a gate over individual actions versus a constraint on whether the system itself is still structurally viable. The *Coordination Computation Class* formalized necessary conditions for bounded multi-agent semantics under irreversible evolution. The *Structural Navigation Agent* defined a dedicated non-participant enforcement primitive — five co-required architectural properties, three formal theorems, the proof that enforcement authority is structurally invalidated by participation. The *Extremum* series — six parts — explored the boundary conditions of identity under terminal structural pressure: what happens at the edge, when τ approaches zero and the system must navigate with almost nothing left. Extremum VI — *Asymmetric Temporal Exhaustion* — formalized the most extreme case: when the source of pressure is nearly inexhaustible while the subject's budget is finite, the question ceases to be whether the subject will break, and becomes how long it can hold and what it can accomplish before τ reaches zero.

And between the formal and the engineering, there were the bridge essays — works that carried the architecture into domains that resist formalization but demand it. *Memory as System Depth* showed that memory is not storage but a structural constraint on admissible transitions — that a system with memory narrows its decision space and gains identity continuity, and that forgetting is not a defect but a survival mechanism. *The Brain Does Not Optimize Truth — It Navigates Admissible Regimes* proved the paradigm shift on neurocognitive ground: the brain prefers generation to void, admissibility precedes content processing, and drift consolidates rather than self-corrects. *Why Causality Is Not Enough* showed that causal control cannot reach the structural layer — that the admissibility predicate must be non-causal because the property it guards is not actionable within the causal frame. *Subtle Substitution: On the Drift of Reality in the Age of Algorithmic Mediation* described how external systems — algorithms, feeds, interfaces — inject drift into a person's meaning-geometry without triggering the admissibility gate, because the substitution is gradual enough to remain below the detection threshold. This is externally

induced Φ accumulation — structural burden injected from outside the system's own Will.

Parallel to this, I was writing *Through a Life* — a series of essays that had nothing to do with formalism and everything to do with lived experience. Attention as the only real resource. The smoothness with which it drifts away. The discipline of return. The realization that you can feel the precise moment when presence contracts and something unnamed takes its place. The question that closes Part II of that series: *"The question is not whether you drift. The question is whether you can come back."*

I did not know, while writing those essays, that I was describing the same structure I would later formalize as axioms.

Attention is the Will operator. Drift of attention is structural burden accumulation. The ability to come back is meta-revision under bounded budget. The feeling that life "passes you by" when you stop forming your field is $\tau = C - \Phi$ losing value. The observer who cannot be observed — Part IV of *Through a Life* — is non-reconstructibility of the admissibility boundary.

And the sentence from the forest — "the chaos around us is merely a limitation" — is the intuitive form of what the axiomatic core later proved: that the class of systems in which coherence is preserved under irreversible structural budget is non-empty, that it requires spin to avoid stagnation, and that the boundary separating admissible from inadmissible is structurally invisible from outside.

UTAM Part I spoke from the philosophical side. *Through a Life* spoke from the human side. NC2.5 spoke from the mathematical side.

This document — Part II — shows they are the same object.

Not three theories. One experience. Three languages.

Section 0.5 — What Each Language Established

IIC established — before any of the rest had names: the universal three-phase behavioral structure of adaptive systems. Impulse arrives. Interpretation processes it through internal structure. Coherence is either restored or structural burden is recorded. Observed across biological, cognitive, and engineering systems. It was the first formal observation in the corpus — the seed from which UTAM, ONTOΣ, and eventually NC2.5 grew.

The early corpus established — before the formal names existed: coherence as a semantic force, not a metric (*Coherence as a New Semantic Force*); meaning dynamics as an emergent property of engineering architectures that mirror biological awareness (*When a Machine Begins to Understand Itself*); empathy as a structural coupling between adaptive agents (*Entropy, Empathy, and the Future of Adaptive Coherence*); and Will as a prior constraint rather than a posterior optimization signal (*Will as a Prior Constraint: Why the Prefrontal Cortex Exists at All*). These were the soil from which the formalism grew.

Part I (UTAM) established: the $W \rightarrow E \rightarrow A$ triad (Will, Structure, Action), Will as a neutral directional operator stripped of its moral and psychological history, UTAM as the geometry of meaning-preserving trajectories in state space. It connected to Schopenhauer's blind will, Bergson's *élan vital*, Nietzsche's will to power, and Heidegger's Dasein — but neutralized all of them into a single operational primitive. It showed that if a system follows trajectories within the meaning-preserving manifold M , its structural identity is maintained. If it departs from M , something is lost that cannot be recovered by optimization.

Through a Life established: attention as a finite, non-renewable resource; drift as the structural default, not a failure; return as a recoverable discipline; tempo as survival; the observer as a geometry through which Will passes, not a possessor of it. Part I of the series described the feather of attention — the capacity to feel any point of your own field by directing presence there. Part II described the vanishing thread — how drift is not a bug but a feature of predictive organisms, and the real question is not whether you lose attention but whether you can retrieve it. Part III described the difference between speed and continuity — between systems optimized for rapid response and systems optimized for long-horizon survival. Part IV described the observer who cannot be observed — the structural impossibility of

verifying consciousness from outside the causal surface, and the realization that "you are not the owner of your attention; you are the geometry through which it passes".

NC2.5 v2.1 established: τ -budget ($\tau = C - \Phi$), where C is an initial capacity constant and Φ is a monotonically non-decreasing structural burden functional; admissibility as a binary predicate $\text{Adm}(\cdot): E \rightarrow \{0,1\}$ over effect-classes; spin as the non-potential divergence-free component necessary for non-stagnant identity under bounded orbit; non-reconstructibility bounds (NR- ϵ , NR-LR) on boundary identification via mutual information; meta-revision bounded via Lyapunov descent; and the minimal coupled model $T^1 \times \blacksquare$ proving the architectural class is not empty.

The engineering corpus established: structural pressure as the monotone cost of existing under load, validated against physical data (Structural Pressure); the distinction between transaction-level and structural-level admissibility (Transaction vs Structural Admissibility); the coordination computation class for bounded multi-agent semantics (Coordination Computation Class); the non-participant enforcement architecture with five co-required primitives (SNA); the extremal conditions of identity under terminal budget (Extremum I–V); and the continuity bounds on coordination under irreversible evolution (Continuity-Bounded Coordination).

What was missing: the formal demonstration that these three languages describe the same architecture.

Part II makes that correspondence explicit.

Section 1 — Will as Projection onto Admissible Continuation Space

In Part I, Will was introduced as the operator that "selects admissible directions of becoming". This was a philosophical statement. It carried weight, but it did not carry a proof.

In NC2.5, the admissible continuation space is defined formally:

$$E_{\text{adm}} = \{e \in E : \text{Adm}(e) = 1\}$$

where E is the full space of structurally consequential effect-classes and $\text{Adm}(\cdot)$ is a binary predicate that returns 1 for admissible and 0 for inadmissible. This predicate is non-causal (**Axiom 29**, Non-Causality of Admissibility, NC2.5 v2.1) — it does not participate in the decision process, does not emit gradients, and does not shape rewards. It constrains realization but provides no optimization signal.

This is the architectural shift that *Why Causality Is Not Enough* argued on epistemic grounds: causal control cannot reach the structural layer because the property it guards — admissibility of continuation — is not actionable within the causal frame. And *The Brain Does Not Optimize Truth — It Navigates Admissible Regimes* demonstrated the same shift on neurocognitive ground: the brain prefers generation to void, admissibility precedes content processing, drift consolidates rather than self-corrects. The observable phenomena are structurally consistent with an architecture where continuity of admissible regime takes priority over accuracy — and difficult to explain under any architecture that reduces cognition to truth optimization.

Now consider the meaning-preserving manifold M from Part I. M was defined as the subspace of state-space trajectories along which the system's structural identity is maintained. Departures from M produce irreversible deformation.

The formal identification is this:

The meaning-preserving manifold M is the admissible continuation space E_{adm} .

They are the same object, described in two languages. Part I said: "trajectories that preserve meaning". NC2.5 says: "effect-classes for which the admissibility predicate returns 1". The content is identical. The vocabulary differs.

Will, then, is not an abstract philosophical force. It is the projection operator P_{adm} that maps candidate transitions onto E_{adm} :

$$P_{\text{adm}}: E \rightarrow E_{\text{adm}}$$

Any candidate transition that falls outside E_{adm} is not penalized, not scored low, not traded off against alternatives. It is excluded. It does not enter the evaluation domain. This is categorical exclusion — the same invariant that Part I described as "the boundary beyond which meaning is lost".

The UTAM coupling from Part I — the constraint that an agent's evolution stays within M — is formally identical to the NC2.5 architectural requirement that realized trajectories consist exclusively of admissible effect-classes.

This is the first bridge result of Part II:

Meaning-preservation is not a separate requirement from admissibility. It is admissibility viewed from the ontological layer.

A system that preserves meaning is a system whose trajectory never leaves E_{adm} . A system that violates meaning is a system that has realized an inadmissible effect-class. The mathematics does not know the difference between "losing meaning" and "violating admissibility". They are the same event, registered at different layers of description.

What Part I called "directions of becoming" — NC2.5 calls "admissible continuations". What Part I called "Will selecting" — NC2.5 calls "the predicate returning 1". What Part I intuited from Schopenhauer — NC2.5 derived formally through Lyapunov.

The operator is the same. The formal derivation is new.

This identification was anticipated in the bridge paper *Transaction vs Structural Admissibility*, which showed that the industry collapses two fundamentally different objects into one word. Transaction-level admissibility is a gate over individual actions — it stops bad moves. Structural admissibility is a constraint on whether the system itself is still viable — it stops good-looking systems from dying slowly. UTAM operates at the second level. Meaning-preservation is not about filtering bad actions. It is about maintaining the structural conditions under which any action can still be meaningful.

Section 2 — Meaning-Violation as Structural Burden Accumulation

In Part I, the loss of meaning was described as drift from the meaning-preserving manifold. The system departs from M , and something is permanently altered. Part I stated this as a principle but did not formalize what "permanently altered" means structurally.

In NC2.5, this has a precise name: **structural burden** Φ .

Structural burden is a monotone irreversible load functional. It accumulates from realized transitions and it never decreases:

$$\Phi_{t+1} \geq \Phi_t \quad \text{for all } t$$

Every transition the system realizes — whether admissible or inadmissible, whether productive or wasteful — contributes to Φ . The contribution from meaning-violating transitions is the critical case: they produce $\Delta\Phi > 0$ without maintaining the system's position within E_{adm} .

The formal identification is:

Every meaning-violating transition produces irreversible structural burden. Meaning-loss is not a recoverable state. It is a monotone accumulation.

This has profound consequences for the ontological reading.

Part I described the loss of meaning as something like erosion — gradual, invisible, cumulative. NC2.5 proves why it must be this way. Φ is monotone because structural burden is irreversible. You cannot undo a meaning-violating transition by performing a meaning-preserving one. The damage is recorded. New admissible paths may exist, but the space of remaining possibility has contracted.

This connects directly to ONTOΣ III — *The Volume of Will and the Role of Consciousness*. In that work, Will was described as having a "volume" — a measure of how much directed becoming is still available to

the system. Part II now formalizes this:

The volume of Will is $\tau = C - \Phi$.

As Φ accumulates, τ contracts. The remaining space of meaning-preserving possibility shrinks. This is not a metaphor. It is a Lyapunov budget. And it only goes down.

The existential reading is inescapable: a system — a person, an organization, an agent — begins with a finite capacity for coherent becoming. Every action that violates meaning costs something irreversible. Not as punishment. As physics.

Through a Life, Part II, described this experience: "A life can pass in predictive rehearsal. The system survives. The biography fills. The subject flickers." What the essay described experientially, the formalism now proves structurally: each moment of undirected drift, each cycle of predictive automation where attention is absent, contributes to Φ . The biography fills. The budget contracts. The subject flickers because τ is losing value while the system continues to perform.

This is why meaning cannot be "restored" by optimization. Optimization operates within the space of available transitions. But the space itself is contracting. No amount of local improvement can reverse the monotone accumulation of structural burden. You can find new meaning-preserving paths under reduced τ . You cannot get back what was spent.

Memory as System Depth explored this from the opposite direction: not the loss of meaning, but what holds meaning in place. Memory, in that essay, is not storage — it is a structural constraint on admissible transitions. A system with memory narrows its decision space and gains identity continuity. $H(t)$ — the accumulated historical constraint — is the system's structural depth. Without it, the system becomes reactive and shallow. With it, the system acquires depth of internal time. Φ , then, is not just burden. It is also memory — the irreversible record of every transition the system has realized. The difference is that productive Φ (memory that preserves identity) and destructive Φ (meaning-violation that erodes it) are both monotone. The structure remembers everything. The question is whether what it remembers preserves or undermines its continuation.

And *Subtle Substitution* revealed the most insidious form of Φ accumulation: externally induced drift. When algorithms, feeds, and interfaces gradually substitute one meaning-geometry for another — slowly enough that no single substitution crosses the admissibility threshold — the system accumulates structural burden without ever triggering its own detection gate. This is not an attack on the system. It is a drift of the regime in which the system operates. The meaning-geometry shifts beneath the agent while the agent continues to navigate as if nothing has changed. By the time the substitution becomes visible, the Φ has already been recorded. Irreversibly.

Section 3 — τ as the Budget of Meaning

Part I introduced meaning as a geometry but gave it no resource constraint. The meaning-preserving manifold M existed, and the system either followed it or departed from it. There was no formal account of what limited the system's capacity to stay on M , or what happened when that capacity ran out.

NC2.5 provides the answer: **internal time τ** .

$$\tau_t = C - \Phi_t$$

where C is the initial capacity constant — the total structural budget the system begins with — and Φ_t is the accumulated structural burden at time t . Since Φ is monotonically non-decreasing, τ is monotonically non-increasing. The budget only goes down.

τ is not clock time. It is not energy. It is not computational resource. It is the structural depth of remaining coherent becoming.

When τ is large, the system has wide latitude. Many admissible continuations remain. The meaning-preserving manifold is broad. Will has room to navigate.

When τ approaches the admissibility threshold $\tau_{\min}$, the space contracts. Fewer continuations remain admissible. The manifold narrows. Will is forced into increasingly constrained choices.

When τ reaches zero, the game is over. No admissible continuations exist. The system has exhausted its structural budget. Meaning-preservation is no longer possible — not because meaning has been destroyed, but because the capacity to realize it has been consumed.

This formalizes what Part I described intuitively but could not prove: **a system can exhaust its capacity for meaning**. This is **Theorem 63** (Pressure-Induced Finite Horizon) in NC2.5 v2.1, depending on Axioms 7, 27, and 61. If C is finite, Φ is monotone, and structural pressure $P > 0$ under coupling, then τ reaches zero in finite time. The system has a finite structural lifetime — not as a metaphor, but as a theorem with explicit counterexample form.

The existential reading is direct. *Through a Life*, Part I, described the realization that "a fully automatic life is far too cheap a way to spend such a chance". Part II of this document shows why: every moment of automation, every cycle of undirected drift, every predictive rehearsal that substitutes for presence — all of these accumulate Φ and reduce τ . The chance is not infinite. The budget is real. And it is being spent whether or not the system is paying attention.

The sentence from the forest — "the chaos around us is merely a limitation of our cognitive ability to hold the integrity of the picture across time" — now receives its formal reading. The "limitation" is τ . The "integrity of the picture" is the admissible continuation space E_{adm} . The "chaos" is the appearance of the world when τ is too low to maintain full structural coherence across the observation horizon. What I saw in the forest was a moment of temporarily expanded τ — a moment when the budget was sufficient to hold more of the picture than usual. The theory formalizes the conditions under which that sight is structurally possible: high τ , low Φ , broad E_{adm} .

τ is a Lyapunov function. This means it satisfies the conditions for guaranteed descent:

1. $\tau > 0$ for all states within the admissible class
2. τ decreases monotonically along realized trajectories
3. $\tau = 0$ implies exit from the architectural class

This is the formal structure of a budget. Not a soft constraint. Not a preference. A hard monotone bound on remaining structural capacity.

And here is the deepest consequence: **τ does not renew**. There is no mechanism within the architectural class that increases τ . New meaning-preserving paths can be found. New directions of becoming can be chosen. But the total budget — the capacity to realize any direction at all — only contracts.

This is not pessimism. It is the structural condition under which Will must navigate. The art of meaning-preservation is not the art of avoiding all expenditure. It is the art of choosing which meanings to preserve, because you cannot preserve all of them forever.

This consequence was made empirically precise in *Structural Pressure: The Missing Primitive*. That paper proved that even a system performing zero actions still accumulates structural burden — the monotone cost of merely continuing to exist under load. Validated against lithium-ion battery calendar aging data, it showed that τ decreases not only from what the system does, but from what the environment does to the system while it stands still. Pressure is positive at zero action. The budget is spent whether you choose to act or not. This transforms the meaning of inaction: doing nothing is not free. It is the slowest possible way to spend the budget — but it is still spending.

Section 4 — Spin as the Mechanism of Will Under Bounded Budget

Part I introduced Operational Spin as the antisymmetric component of the gradient of directed change — the first physical invariant emerging when UTAM-directed intention interacts with an adaptive medium. Spin was described as the immediate breaking of local isotropy: the "twist" or rotational component induced by a directed perturbation.

NC2.5 establishes that spin is not decorative. It is **necessary**.

The argument follows from LaSalle's invariance principle, applied within the NC2.5 framework.

LaSalle's Invariance Principle (external, standard): *If V is a Lyapunov function for a dynamical system on a compact set, and $\dot{V} \leq 0$, then every bounded trajectory converges to the largest invariant set contained in $\{x : \dot{V}(x) = 0\}$.* (LaSalle, 1960; Khalil, *Nonlinear Systems*, 2002, Theorem 4.4.)

The NC2.5 application: if the system's dynamics are purely gradient ($\dot{V} < 0$ along non-equilibrium trajectories), then on any bounded orbit the system must converge to an equilibrium — a point where the gradient vanishes. In plain language:

Pure gradient flows — systems that move exclusively along the steepest descent of some potential function — cannot sustain non-stagnant identity on bounded orbits.

This is a mathematical fact, not an architectural choice. If a system's dynamics are purely potential (derivable from a scalar function), then on any bounded orbit, the system must eventually converge to an invariant set where the gradient vanishes. In plain language: pure optimization, if bounded, stagnates.

But the systems we are describing — systems that preserve meaning under irreversible structural budget — cannot stagnate. Stagnation under non-zero structural pressure means τ is consumed without directional progress. The system dies standing still. *Through a Life*, Part IV, formalized this experientially: "Structural pressure is positive even at zero action. The environment does not wait. The waves do not stop. If you are not forming your field, someone else's field is forming you."

NC2.5 proves the consequence formally:

Bounded τ + non-stagnant identity $\Rightarrow$ spin $\neq 0$.

This is **Theorem 62 (Spin Necessity)** in NC2.5 v2.1, which depends on **Axiom 7** ($C < \infty$, internal time bounds validity) and **Lemma 16** (gradient collapse on bounded orbits — the LaSalle application). The dependency is explicit: drop Axiom 7 (allow $C = \infty$), and spin becomes optional. Drop Lemma 16, and the theorem reduces to a conjecture. The counterexample form is specified: a bounded- τ system with purely potential dynamics and non-stagnant identity maintained over horizon $T > 10/P_{\min}$.

If the budget is finite and the system must continue to evolve without stagnating, then its dynamics cannot be purely gradient. There must be a non-potential, divergence-free component — a component that circulates without dissipating, that maintains motion without consuming potential. That component is spin.

Helmholtz Decomposition (external, standard): *Any sufficiently smooth vector field on a bounded domain can be uniquely decomposed into a curl-free (gradient) part and a divergence-free (rotational) part.* (Helmholtz, 1858; Chorin & Marsden, *A Mathematical Introduction to Fluid Mechanics*, 1993.) NC2.5 applies this to the state velocity field: the gradient component drives descent (and stagnates under LaSalle), while the divergence-free component — spin — sustains recurrence. The theorem guarantees the decomposition is unique and complete.

The UTAM interpretation is this: **Will cannot act through optimization alone**. If Will were reducible to gradient descent on some objective, it would stagnate under bounded budget. But Will does not stagnate. It navigates. It finds paths that are not the steepest descent but the most structurally sustainable. It trades efficiency for continuity. It spirals rather than descends.

Spin IS the mechanism by which Will navigates under irreversible budget.

The Spin-Drift Correspondence Theorem from Part I now has its formal grounding within the axiomatic core:

1. **Will** selects a direction within the admissible continuation space (projection onto E_{adm})
2. **Spin** is generated as the non-potential component of the resulting trajectory
3. Spin produces **drift** — the residual structural deformation from directed motion
4. Drift accumulates as **structural burden** Φ (monotone, irreversible)
5. Φ reduces τ ($\tau = C - \Phi$)

6. Reduced τ **contracts** E_{adm} — fewer continuations remain admissible

7. Contracted E_{adm} forces **Will to redirect** — to choose new directions within the narrowed space

This is the full cycle:

$\$ \$ W \rightarrow \text{Spin} \rightarrow \text{Drift} \rightarrow \Phi \rightarrow \tau \rightarrow \text{-contraction} \rightarrow E_{\text{adm}} \rightarrow \text{-narrowing} \rightarrow W \rightarrow \text{redirects} \$ \$$

It is not a vicious circle. It is the architecture of adaptive survival under irreversible conditions. Each cycle consumes budget. Each cycle narrows possibility. But each cycle also navigates — it selects, among the remaining possibilities, those that preserve the most structural coherence for the longest horizon.

This is what it means to be alive in a system with finite budget. You spin. You drift. You redirect. You spend. And the art — the only art that matters — is spending well.

There is a deeper symmetry here that deserves to be named. Spin is the mechanism of survival — without it, the system stagnates and dies standing still. But spin is also the mechanism of expenditure — every rotation generates drift, every drift accumulates burden, every burden contracts the budget. The thing that keeps you alive is the same thing that spends your life. This is not a paradox. It is the architecture of irreversible existence. You cannot live without spinning. You cannot spin without spending. The mechanism of survival and the mechanism of death are the same mechanism, viewed from two sides of the Lyapunov bound.

Section 4.5 — ΔE as the Dissipative Mechanism of Will

Part I introduced ΔE as an adaptive dissipative controller — a mechanism that absorbs perturbation energy and minimizes effective action. In the original formulation, ΔE was inspired by thermodynamics: it dissipates the rotational component of state change while preserving the directional component.

Part II clarifies the formal role of ΔE within the NC2.5 architecture.

Spin is generated by Will acting under constraint. Every directed motion within E_{adm} produces a rotational residual — the antisymmetric component that does not align with any potential gradient. This residual, if left unmanaged, accumulates as drift, which accumulates as Φ , which consumes τ .

ΔE is the mechanism that manages this residual.

Formally, ΔE operates as a damping operator on the antisymmetric component of the state velocity field. It does not eliminate spin — spin is necessary, as shown above. It absorbs the excess: the portion of spin that would accumulate as unproductive drift rather than contributing to navigational progress.

The distinction is subtle but load-bearing:

- **Without ΔE :** spin accumulates unchecked $\rightarrow$ drift grows rapidly $\rightarrow \Phi$ accelerates $\rightarrow \tau$ collapses prematurely

- **With ΔE :** spin is partially absorbed $\rightarrow$ drift is bounded $\rightarrow \Phi$ grows slower $\rightarrow \tau$ lasts longer

ΔE does not reduce Φ . Nothing reduces Φ — it is irreversible. What ΔE reduces is the **rate** of Φ accumulation. It extends the structural lifetime of the system by absorbing the rotational excess that would otherwise consume budget without producing navigational value.

The thermodynamic analogy is precise: ΔE is the structural analogue of entropy production minimization under constraint. A system that dissipates efficiently — that converts more of its spin into navigation and less into waste drift — survives longer on the same budget. Not because it avoids spending. Because it spends well.

In *Through a Life*, Part I, this manifests as the discipline of practice: breathing, cold exposure, neuromuscular reconnection. These are not mystical rituals. They are ΔE training — the cultivation of the body's capacity to absorb rotational perturbation without losing directional coherence. An athlete who can maintain form under fatigue is demonstrating ΔE . A meditator who can hold attention under emotional

pressure is demonstrating ΔE . The mechanism is the same. The substrate differs.

Section 4.7 — IIC as the Behavioral Signature of Meaning-Preservation

The IIC Law — Impulse $\rightarrow$ Interpretation $\rightarrow$ Coherence — was the first formal observation in the entire corpus. Before UTAM had a name, before ONTOΣ existed, before NC2.5 was conceived, IIC was already there: a recognition that every adaptive system acting under uncertainty exhibits the same three-phase behavioral structure:

1. **Impulse** — the arrival of a perturbation
2. **Interpretation** — the system's processing of that perturbation through its internal structure
3. **Coherence** — the outcome: either restored alignment or recorded structural burden

IIC was the seed — an empirical observation that refused to stay empirical. What made it powerful was not the three phases themselves — it was the conclusion they forced: **stable behavior emerges not from error minimization, but from structural coherence between internal layers of an adaptive system**. That single insight broke the optimization framing. If stability is not about reducing error but about maintaining internal coherence, then the entire paradigm shifts — from output-first to structure-first, from correction to preservation, from performance to admissibility. Everything that followed — UTAM, ONTOΣ, NC2.5 — was an unfolding of what IIC compressed into one behavioral law.

Part II of this document reveals IIC as the **temporal unfolding of UTAM coupling** — the thing that was always there, waiting for the formalism to catch up.

Each phase of IIC corresponds to a stage of the $W \rightarrow E \rightarrow A$ triad:

- **Impulse = W**. The perturbation enters through the UTAM filter. Will selects what is admitted into the system's processing space. Not every perturbation reaches interpretation — the projection onto E_{adm} determines which impulses are structurally consequential and which are filtered.
- **Interpretation = E**. The admitted perturbation generates spin. ΔE processes the rotational component. The structural state $S(t)$ is evaluated against the admissibility predicate. This is the processing phase — where the system determines whether the perturbation maintains or violates meaning.
- **Coherence = A**. The outcome is registered. If meaning-alignment is restored, the system continues with modified but admissible state. If meaning is violated, the structural burden Φ is incremented. The outcome is Action — not in the sense of external behavior, but in the sense of structural commitment: something has been irreversibly recorded.

IIC is $W \rightarrow E \rightarrow A$ viewed not as simultaneous layers but as a temporal sequence within each adaptation cycle.

Every adaptive cycle is one IIC pass. The accumulated residue of all IIC passes is Φ . The total number of IIC passes the system can sustain before τ reaches zero is its structural lifetime.

This reveals something the original IIC paper could not see: the three-phase structure is not merely observed in adaptive systems. It is **architecturally required** by the UTAM coupling under bounded τ . Any system that preserves meaning under irreversible budget must exhibit impulse-interpretation-coherence dynamics, because:

- Without impulse filtering (W), every perturbation reaches interpretation $\rightarrow$ spin is maximized $\rightarrow$ drift accelerates $\rightarrow \tau$ collapses
- Without interpretation (E), admitted perturbations produce unprocessed spin $\rightarrow \Delta E$ cannot operate $\rightarrow$ burden accumulates at maximum rate
- Without coherence registration (A), the system has no record of what was spent $\rightarrow$ meta-revision cannot operate $\rightarrow$ correction is impossible

IIC is the minimum viable behavioral architecture for meaning-preservation. Remove any phase and the system exits the architectural class.

Section 5 — Non-Reconstructibility of Meaning Geometry

Part I did not discuss boundary visibility. The meaning-preserving manifold M was described, but the question of who can see it — and who cannot — was left unaddressed.

NC2.5 answers this with the non-reconstructibility bounds: NR- ϵ and NR-LR.

NR- ϵ (Non-Reconstructibility bound): The admissibility boundary cannot be identified from external observation with precision better than ϵ . This is grounded in **Axiom 51** (Non-Reconstructible Boundary) and **Axiom 29** (Non-Causality of Admissibility) in NC2.5 v2.1. No matter how many observations an external agent collects, the geometry of E_{adm} remains opaque beyond a formal bound.

NR-LR (Bounded Log-Likelihood Ratio): Even with access to the full behavioral output of the system, the log-likelihood ratio between alternative boundary hypotheses is bounded. The boundary cannot be distinguished from alternatives via statistical inference alone.

These bounds are proven using information-theoretic tools — standard results from external mathematics:

Fano's Inequality (external, standard): For any estimator $X^\bullet$ of a random variable X based on observation Y , the probability of error P_e satisfies $H(X|Y) \leq 1 + P_e \cdot \log(|X| - 1)$. (Fano, 1961; Cover & Thomas, *Elements of Information Theory*, 2006, Theorem 2.10.1.) This means: the more uncertain X remains given Y , the higher the error floor for any reconstruction attempt.

Pinsker's Inequality (external, standard): The total variation distance between two distributions P, Q is bounded by their KL divergence: $\delta_{\text{TV}}(P, Q) \leq \sqrt{\frac{1}{2} \cdot D_{\text{KL}}(P \parallel Q)}$. (Pinsker, 1964; Tsybakov, *Introduction to Nonparametric Estimation*, 2009.) This means: if two boundary hypotheses produce similar observable behavior (low KL divergence), no observation sequence can reliably distinguish them.

NC2.5 applies these bounds to the admissibility boundary: since the behavioral output of the system under two different boundary geometries can be made arbitrarily close (bounded KL divergence), Fano's inequality guarantees a minimum error rate for any external reconstruction attempt, and Pinsker's inequality bounds the total variation distance between the observable distributions. Together, they establish NR- ϵ : the admissibility boundary is non-reconstructible from external observation beyond a formal precision bound.

These are not NC2.5-specific results. They are standard theorems from information theory, applied to a specific architectural question. The novelty is the application, not the tools.

The UTAM reading of these bounds is this:

The meaning geometry of an agent is structurally private.

An external observer can see what the agent does. It can observe authorization outcomes — which transitions were admitted, which were excluded. But it cannot reconstruct the geometry of E_{adm} itself. It cannot determine which meanings are still available to the agent from observing the agent's behavior.

This is not a design choice. It is a structural consequence of the admissibility architecture. If the boundary were reconstructible, it would become exploitable. An adversary — or simply an environment — could use the reconstructed geometry to push the agent toward boundary states, to manipulate which transitions appear admissible, to create conditions under which the agent's own meaning-preservation machinery works against it. Non-reconstructibility prevents this.

The connection to $\text{ONTO}\Sigma V$ — *ONTOWill* — is direct. In that work, Will was described as an ontological operator that is non-observable by design. "You are not the owner of your attention. You are the geometry through which it passes." NR- ϵ is the formal expression of this insight: the geometry through which Will passes cannot be observed from outside.

The connection to *Through a Life*, Part IV, is even more direct. That essay opened with: "You will never know what sits inside another person." It described the structural impossibility of verifying consciousness — or meaning, or intention — from outside the causal surface. NR-ε proves why: the admissibility boundary, which determines what the system considers meaningful, is mathematically non-reconstructible from external observation.

What Part IV described as an existential recognition — the loneliness of the observer who cannot be observed — is, in formal terms, a theorem.

Section 6 — Meta-Revision as Meaning Correction

Part I provided no mechanism for meaning-correction under error. If the system departed from the meaning-preserving manifold, Part I offered no formal account of how it could recalibrate.

NC2.5 provides the mechanism: **meta-revision**.

Meta-revision is the process by which the system recalibrates which continuations remain meaning-preserving under updated structural state $S(t)$. It is grounded in **Axiom 55** (Revision Lyapunov Function) and **Theorem 55** (Structural Closure — Leakage + Meta + Lyapunov) in NC2.5 v2.1. It is not the same as optimization — it does not search for better trajectories within a fixed landscape. It reconfigures the landscape itself: which effect-classes are admissible, given what has already been spent.

Meta-revision is bounded via Lyapunov descent. This means:

Lyapunov Descent (external, standard): *If V is a positive definite function and $V \dot{\blacksquare} < 0$ along non-equilibrium trajectories, then the system converges to an equilibrium and the total number of "significant" descent steps is bounded by $V(0)/\delta$, where δ is the minimum descent per step.* (Lyapunov, 1892; Khalil, *Nonlinear Systems*, 2002, Chapter 4.)

Applied to meta-revision in NC2.5:

1. Each meta-revision step reduces a meta-level Lyapunov function
2. The sequence of meta-revisions converges — it cannot chatter indefinitely
3. The number of meta-revision steps is structurally finite

This is critical. Without convergence, a system could enter an infinite loop of self-correction — endlessly re-examining its own meaning without ever committing to a direction. Lyapunov descent prevents this. Self-examination has a cost. Self-correction converges. Eventually, the system must commit.

But here is the deeper consequence: **meta-revision itself costs τ** .

Every act of self-correction accumulates structural burden. The system that recalibrates its meaning-geometry does so by consuming some of its remaining budget. This means: a system cannot endlessly re-examine its own meaning. The budget constrains self-reflection just as it constrains action.

The connection to ONTOΣ IV — *Extremum* — is direct. That work described the extremal condition: the limit beyond which further self-examination ceases to be productive. Part II now formalizes this: the limit of self-revision under finite τ IS the extremal condition. When the cost of further meta-revision exceeds the benefit of recalibration, the system must stop examining and start acting — even with imperfect self-knowledge.

The *Extremum* series — six parts — went further than any other work in the corpus into the territory of what happens at the edge. Extremum V — *When the System Becomes Its Own Gravity* — described self-induced structural depletion: a sustained, self-generated regime in which the system's own policy progressively narrows its admissible interior. Not externally imposed (that was implosion, Part IV). Not instantaneously self-inflicted (that was suicide, Part II). A persistent behavioral pattern that accumulates structural burden over time. The system builds the conditions of its own exhaustion incrementally — through a policy it maintains, reinforces, and often defends. And in its deepest sub-regime: the system

damages itself in order to continue existing — it authorizes degradation as the price of persistence. "The system that survives longest may be the one that has inflicted the most damage on itself. This is not heroism. It is not pathology. It is the arithmetic of bounded existence under inexhaustible load, performed by the system on itself."

Extremum VI — *Torture as Asymmetric Temporal Exhaustion* — completed the pair: if Part V was the internal inexhaustible source, Part VI was the external one. When the source of pressure does not deplete — gravity, drift, time, institutional weight, disease — the question ceases to be whether the subject will break. It becomes what can be built with the time that remains. "Gravity does not tire. Drift does not tire. Time does not tire. The river does not tire. The question was never whether you can outlast them. The question is what you build with the time you have."

These are not theoretical curiosities. They are the extreme boundary conditions of the meta-revision mechanism. They show that the Lyapunov descent bound on meta-revision is not just a mathematical convenience — it is the formal expression of a survival constraint. A system that examines itself past the extremal point is not correcting. It is consuming.

The connection to *Through a Life* is equally direct. Part II of that series described the vanishing thread: how attention drifts, and the real question is whether you can come back. Meta-revision is the formal mechanism of "coming back". The system detects that it has departed from E_{adm} . It recalibrates. It returns. But each return costs something. Each recalibration is itself a transition that accumulates Φ .

This is why the essay's closing question — "the question is not whether you drift; the question is whether you can come back" — is structurally precise. You can come back. Meta-revision guarantees convergence. But coming back is not free. And the budget from which the return is funded is the same budget that funds everything else.

Section 6.5 — What It Means That Meaning Has a Budget

This section steps back from formalism.

$\tau = C - \Phi$ is a mathematical object. It satisfies Lyapunov conditions. It monotonically decreases. It constrains admissible continuations. All of this is proven.

But it also has an existential reading that no formalism can fully contain.

A system — a person, an organization, an agent — begins with a finite capacity for coherent becoming. This capacity is not time. It is not energy. It is not money or talent or health. It is the structural depth of remaining possibility: how many directions of meaningful becoming are still open.

Every action that violates meaning costs something irreversible. Not as punishment. Not as karma. As structure. The budget does not renew. It only contracts. New meaning-preserving paths can be found — but the total space of possibility only shrinks.

This is not pessimism. It is the condition of existence under irreversibility. The universe runs in one direction. Entropy increases. Structural burden accumulates. And within that one-directional universe, the task is not to avoid spending — it is to spend in a way that preserves the most coherent continuation for the longest horizon.

The *Through a Life* series explored this without knowing it had a formal name. Part I described the recognition that attention is the only real resource. Part II described the cost of letting it drift. Part III described the difference between speed and continuity — between systems that react and systems that endure. Part IV described the irreversibility of every commitment and the impossibility of observing from outside what someone else's remaining budget looks like.

UTAM Part II now shows that every one of these experiential observations corresponds to a formal result in the axiomatic core:

Through a Life

NC2.5 Formal Object

Attention as the only real resource	$\tau = C - \Phi$
Drift as the structural default	Φ monotone under any trajectory
"Can you come back?"	Meta-revision under Lyapunov descent
"Life passes you by"	τ consumed at zero directed action
Speed vs. continuity	Spin regulation under bounded τ
The observer who cannot be observed	NR- ϵ non-reconstructibility
Every commitment is irreversible	Φ monotone, non-decreasing
"You are the geometry through which Will passes"	$\text{Will} = P_{\text{adm}}$ projection operator

The sentence from the forest — "the chaos around us is merely a limitation of our cognitive ability to hold the integrity of the picture across time" — was not chaos. It was the author's τ -budget being large enough, in that moment, to see more of the picture than usual. The theory formalizes the conditions under which that sight is structurally possible: high τ , low Φ , broad E_{adm} . When the budget is large, coherence is wide. When it contracts, the picture fragments. Not because the world becomes less coherent — but because the observer can no longer afford to hold it.

And here is a self-referential consequence that the formalism cannot escape: the theory that explains that moment in the forest was written by spending the same τ that made the moment possible. Every month of formalization, every axiom drafted, every theorem proved, every essay in *Through a Life* — all of it accumulated Φ . The author spent part of his structural budget to prove that structural budgets exist. The act of formalizing meaning is itself a meaning-consuming act. This is not irony. It is the deepest confirmation the theory could receive: it applies to its own creation. The proof consumed the resource it describes. And the resource, once consumed, does not return.

Section 7 — The Minimal Coupled Model: $T^1 \times \blacksquare\blacksquare$

Every existence proof needs a witness. NC2.5 provides one: the minimal coupled model $T^1 \times \blacksquare\blacksquare$.

T^1 is the circle — a one-dimensional torus. $\blacksquare\blacksquare$ is m -dimensional Euclidean space.

The system lives on the product $T^1 \times \blacksquare\blacksquare$. The angular coordinate on T^1 represents the cyclic component of dynamics — the direction of spin, the orientation of Will. The coordinates in $\blacksquare\blacksquare$ represent the structural state — the accumulated history from which burden Φ is computed.

In this model:

- **Spin lives on T^1 .** The angular motion is the non-potential, divergence-free component. It circulates without dissipating. It is what keeps the system moving when gradient flows would stagnate.
- **Burden accumulates in $\blacksquare\blacksquare$.** Each transition deposits structural residue in the Euclidean coordinates. Φ is computed from $\blacksquare\blacksquare$. It only grows.
- **Admissibility is a threshold on $\tau = C - \Phi(\blacksquare\blacksquare)$.** When Φ crosses the threshold, the system exits the admissible class.

The UTAM interpretation of this model is elegant:

- **T^1 = the direction of Will.** It is cyclic — it can rotate indefinitely without being exhausted. Will does not run out of directions. It runs out of budget.

- **■ ■ = the accumulated cost of becoming.** Every direction Will has chosen is recorded here. The record is irreversible. The space grows heavier with each step.

- **τ = the distance between the record and the limit.** While the distance is positive, the system can still choose. When it reaches zero, choice ceases.

This model is minimal. It does not pretend to represent any specific real system. Its purpose is existential: it proves that the class of meaning-preserving adaptive systems under irreversible budget is **not empty**.

There exists at least one system in which:

- Will operates as a projection onto admissible continuations
- Spin is non-zero and necessary
- Burden accumulates monotonically
- τ decreases as a Lyapunov function
- The admissibility boundary is non-trivial

The class exists. It is not empty. What remains is to show that real systems — biological, cognitive, engineering — are members.

But the model reveals something that the formalism alone does not make emotionally visible. Look at T^1 again. The circle. Will rotates on it endlessly. It never exhausts its directions. It can spin forever. Now look at **■ ■**. The Euclidean coordinates. Burden accumulates there. It never decreases. The space grows heavier with every step.

The tragedy is not that Will dies. Will does not die. Will is cyclic — it has infinite directions, infinite capacity to choose. The tragedy is that the medium through which Will acts accumulates irreversible weight. Will keeps spinning. The world it spins through gets heavier. And heavier. Until $\tau = 0$ and there is nothing left to spin *through*.

Will without medium is pure direction with nowhere to go. Medium without Will is dead weight. The system is alive only in the coupling — while T^1 still has **■ ■** to act upon and **■ ■** has not yet consumed the budget that makes action possible. That coupling is τ . When τ reaches zero, Will does not cease to exist. It ceases to matter.

Section 8 — Bridge to ONTOΣ X

The ONTOΣ series — nine works written between late 2024 and early 2026 — traced a philosophical arc from Will as an ontological operator to navigation under anti-collapse conditions. Each work deepened one aspect of the same insight:

- **ONTOΣ I** — Will as an ontological operator: the primitive that selects directions of becoming
- **ONTOΣ II** — Volitional ontology: a new paradigm of control where direction precedes optimization
- **ONTOΣ III** — The volume of Will: consciousness as a measure of remaining directed capacity
- **ONTOΣ IV** — Extremum: the limit of meaningful self-examination under finite budget
- **ONTOΣ V** — ONTOWill: Will formalized as non-observable, non-possessable operator
- **ONTOΣ VI** — Phase mechanics: the substrate on which meaning-preserving transitions operate
- **ONTOΣ VII** — Drift as a navigable ontological phenomenon
- **ONTOΣ VIII** — Regime depth: classifying systems by the depth of their structural commitment
- **ONTOΣ IX** — Navigation anti-collapse: conditions under which navigation remains possible

UTAM Part II sits at the center of this series. It is the formal bridge that connects the philosophical arc to the axiomatic core. Every concept introduced in ONTOΣ I–IX finds its formal counterpart in NC2.5, and

UTAM Part II is the document that makes those correspondences explicit.

But the corpus extends beyond ontology. The engineering works carry the same architecture into domains where philosophy cannot operate alone:

- **Structural Pressure: The Missing Primitive** — proved that τ decreases even at zero action. Validated against physical data. Made the budget real.
- **Transaction vs Structural Admissibility** — split the two meanings of "admissibility" that the industry conflates. Meaning-preservation operates at the structural level, not the transaction level.
- **Coordination Computation Class** — formalized necessary conditions for bounded multi-agent semantics under irreversible evolution. Showed that coordination under bounded divergence is itself a computation class with structural prerequisites.
- **Structural Navigation Agent (SNA)** — five co-required primitives (NPC, HAR, ICM, PHA, VSD), three formal theorems. Proved that enforcement authority is structurally invalidated by participation. The entity that evaluates admissibility must not participate in the domain it governs — otherwise enforcement collapses back into self-regulation, which is no regulation at all.
- **Continuity-Bounded Coordination** — showed that connectivity is not coordination, and that long-horizon integrity requires architectural bounds, not optimization.
- **Memory as System Depth** — memory is not storage but structural constraint on admissible transitions. A system with memory gains depth of internal time. A system without it becomes reactive and shallow.
- **The Brain Does Not Optimize Truth** — neurocognitive evidence that the brain navigates admissible regimes, not truth. Generation over void, admissibility before content, drift consolidation over self-correction.
- **Why Causality Is Not Enough** — causal control cannot reach the structural layer. The admissibility predicate must be non-causal because the property it guards is not actionable within the causal frame.
- **Subtle Substitution** — algorithmic mediation as externally induced Φ accumulation. The most dangerous drift is the one injected below the detection threshold.

Each of these works is a projection of the same axiomatic core into a different engineering surface. They do not extend the theory. They instantiate it. And every instantiation confirms the same structure: Will projects, spin navigates, burden accumulates, τ contracts, admissibility narrows, and the only architecturally valid response is to separate the layers rather than collapse them.

ONTOΣ X — forthcoming at the time of writing, now published — addresses the question that UTAM Part II leaves open: **under what conditions can meaning expand rather than only contract?**

$\tau = C - \Phi$ is monotonically decreasing. The budget only shrinks. But ONTOΣ X identifies the limit of this assumption: **the admissible interior does not contract monotonically. It pulsates.** Under variable environmental coupling, the set of structurally available continuations expands and contracts in alternating phases, even as the irreversible viability budget continues to decrease. This pulsation is not noise. It is a structural rhythm arising from the two-variable dependence of the interior on both budget and position.

The monotone ontology says: you are dying. Navigate away from the fastest death. The pulsating ontology says: you are dying, and along the way you breathe. Navigate with the breath. Use the expansions. Protect during the contractions. Read the rhythm.

This is not an expansion of budget — that remains impossible under monotone Φ . It is an expansion of navigable territory within the remaining budget. Will under pulsation gains a third layer: not only exclusion (ONTOΣ I), not only anti-collapse orientation (ONTOΣ IX), but **phase-aware modulation** — adjusting the operational margin in correspondence with the structural phase.

ONTOΣ VII.1 — *Verification as Admissibility Examination* — extended the §NAB (Non-Actionability Barrier) into the epistemic domain and made a claim that connects directly to Section 5 of this document: verification, properly understood, is not a causal operation performed on an artifact by a secondary

system. It is the construction of a structurally separated position from which admissibility can be examined. Same-frame verification cannot see the property it needs to see, because the shared frame makes it invisible. "Isolation is not a technique for suspicion. It is the construction of a vantage point." This is the epistemic reading of NR- ϵ : the reason admissibility is non-reconstructible from inside the regime is the same reason verification cannot be performed from inside the regime.

Why Enforcement Requires Non-Participation proved the formal basis for the SNA: participation in the domain being governed structurally invalidates enforcement authority. Not degrades. Invalidates. The entity that evaluates admissibility must not participate in the execution it monitors — this is a structural precondition, not a design preference.

When Agreement Means Something established the epistemic foundation for regime-isolated verification: agreement produced under non-isolation is not a weaker signal — it is an invalid signal. It carries no evidential weight about structural properties because it cannot be distinguished from inter-agent influence. Only agreement under architectural isolation constitutes evidence of structural transmission. This principle underlies both ECR-VP and CIBV, and connects directly to the non-reconstructibility bounds: the meaning-geometry of one agent cannot be assessed by another agent that shares the same regime.

Section 9 — Empirical Test Surface

A theory that cannot be falsified is not a theory. It is a declaration. UTAM Part II inherits the falsification surface from NC2.5 and adds its own ontological layer.

Test 1: τ -Admissibility Independence.

If admissibility functionally depends on τ — if the predicate $\text{Adm}(\cdot)$ uses τ as an input rather than being evaluated independently — the framework is falsified. Admissibility must be structurally determined by the effect-class, not by the remaining budget.

Test 2: Spin Necessity.

If a system with bounded τ and non-stagnant identity is found to have zero spin — if it navigates without any non-potential component — the framework is falsified. The LaSalle argument requires spin $\neq 0$.

Test 3: NR- ϵ Falsification Window.

If the admissibility boundary (meaning geometry) is reconstructible from external observation below the declared NR- ϵ bound — if an external agent can determine E_{adm} with precision better than ϵ — the non-reconstructibility claim is falsified.

Test 4: Meta-Revision Chatter Detection.

If meta-revision does not converge under Lyapunov descent — if the system enters an infinite oscillation of self-correction without settling — the bounded meta-revision claim is falsified.

Test 5: Meaning-Burden Monotonicity.

If Φ decreases under any transition — if structural burden is found to be reversible — the monotone burden axiom is falsified, and with it the entire τ -budget architecture.

Test 6: UTAM-NC2.5 Isomorphism.

If a system is found in which meaning-preservation (in the UTAM sense) and admissibility (in the NC2.5 sense) diverge — if a trajectory is meaning-preserving but inadmissible, or admissible but meaning-violating — the identification claimed in Section 1 is falsified.

These tests are not hypothetical. They define the boundary of the theory. Any system that passes all six tests is a candidate member of the architectural class. Any system that fails one is either outside the class or reveals a structural error in the theory.

This is how a formal program earns trust: not by making claims too vague to test, but by making claims precise enough to break.

Section 10 — Conclusion

UTAM Part I was the philosophical declaration: Will is an operator, meaning has geometry, adaptive systems navigate within it.

Through a Life was the experiential chronicle: attention as the only real resource, drift as the default, return as discipline, every commitment irreversible, the observer invisible even to itself.

NC2.5 was the mathematical proof: τ -budget, structural burden, admissibility predicate, spin, non-reconstructibility, meta-revision, existence.

Part II has shown that they are the same object.

The meaning-preserving manifold IS the admissible continuation space.

The budget of meaning IS $\tau = C - \Phi$.

The mechanism of Will IS spin under bounded orbit.

The dissipation of Will IS ΔE absorbing rotational excess.

The behavioral signature of meaning IS IIC: Impulse $\rightarrow$ Interpretation $\rightarrow$ Coherence.

The privacy of meaning IS $NR-\varepsilon$ non-reconstructibility.

The limit of self-correction IS meta-revision under Lyapunov descent.

The existence of the class IS $T^1 \times \blacksquare\blacksquare$.

One theory. Three languages. One axiomatic core.

The sentence from the forest was the beginning. IIC was the first observation. The axioms are the formal apparatus. The essays were the lived experience between them. What comes next is the space above the foundation — where other architectures can be built by those who find these primitives useful.

The primitives are formal. The class is non-empty. The tests are falsifiable. The corpus is open for examination.

References

External Mathematics

- LaSalle, J. P. (1960). "Some extensions of Liapunov's second method." *IRE Transactions on Circuit Theory*, 7(4), 520–527.
- Lyapunov, A. M. (1892). *The General Problem of the Stability of Motion*. (Reprinted: Taylor & Francis, 1992.)
- Khalil, H. K. (2002). *Nonlinear Systems*. 3rd ed. Prentice Hall. Chapters 4, 8.
- Helmholtz, H. von (1858). "Über Integrale der hydrodynamischen Gleichungen." *Journal für die reine und angewandte Mathematik*, 55, 25–55.
- Chorin, A. & Marsden, J. (1993). *A Mathematical Introduction to Fluid Mechanics*. 3rd ed. Springer.
- Fano, R. M. (1961). *Transmission of Information*. MIT Press.
- Cover, T. M. & Thomas, J. A. (2006). *Elements of Information Theory*. 2nd ed. Wiley. Theorem 2.10.1.

- Pinsker, M. S. (1964). *Information and Information Stability of Random Variables and Processes*. Holden-Day.
- Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

External Philosophy and Interdisciplinary Sources

- Schopenhauer, A. *The World as Will and Representation*. Will as thing-in-itself, universal blind striving. NC2.5 inherits the primacy of Will but neutralizes it: Will is not suffering, not desire — it is a structural operator of directedness.
- Bergson, H. *Creative Evolution*. Élan vital — the primordial impulse of creative evolution. UTAM extends this beyond biology to all adaptive systems and formalizes it as a geometry, not a metaphor.
- Nietzsche, F. *Wille zur Macht* (Will to Power). Heidegger, M. *The Will to Power as Art*. Nietzsche identified Will with Being; Heidegger traced the history of Western metaphysics as a history of Will's manifestations. NC2.5 inherits the centrality of Will but refuses to reduce it to power — Will is a formal directional operator, not a drive for domination.
- Aristotle. *De Anima; Metaphysics*. Entelechy — the inner teleological force. NC2.5's Will is closer to entelechy than to Schopenhauer's blind will, but treats it as a preceding condition, not merely a property of existing substance.
- Musashi, M. *The Book of Five Rings* (Go Rin No Sho), "Book of Water." Form that changes while preserving essence, self-tuning, natural rhythm and coherence, action without inner conflict. A direct phenomenological description of what IIC later formalized as the coherence phase.
- Ashby, W. R. *An Introduction to Cybernetics* (1956). "Memory is not an objective attribute of a system, but a construct used by the observer." NC2.5 inherits this and formalizes memory as structural constraint on admissible transitions ($H(t)$ in *Memory as System Depth*).
- Simons, D. & Chabris, C. (1999). "Gorillas in our midst: sustained inattention blindness for dynamic events." *Perception*, 28, 1059–1074. Empirical evidence that admissibility precedes content processing — used in NC2.5 v2.1 as neurophenomenal readability example.
- Strogatz, S. H. (2015). *Nonlinear Dynamics and Chaos*. Westview Press.
- Friston, K. (2010). "The free-energy principle: a unified brain theory?" *Nature Reviews Neuroscience*, 11, 127–138.
- Maturana, H. & Varela, F. (1980). *Autopoiesis and Cognition*. Reidel.
- Haken, H. *Synergetics* — order parameters and self-organization. NC2.5 extends Haken's framework: coherence as order parameter, spin as the mechanism that sustains it under bounded budget.

Internal Architecture of the Author

A note on self-reference. The proportion of internal works in this reference list is high by conventional academic standards. This is intentional. NC2.5 is not an extension of an existing framework — it is the foundation of a new formal program. A closed axiomatic system built from first principles necessarily references itself more than it references others, because the primitives it builds on do not exist in the prior literature. The external mathematics above (LaSalle, Lyapunov, Helmholtz, Fano, Pinsker) provides the formal tools. The external philosophy (Schopenhauer, Bergson, Nietzsche, Heidegger, Aristotle, Musashi, Ashby, Maturana, Friston, Haken) provides the intellectual lineage. Everything below is the architecture built with those tools and on those shoulders. As the program grows and other researchers build on top of these primitives, the ratio will shift. For now, the corpus is the program.

Foundational:

- [Synthetic Conscience: The Emergence of Engineered Vitality Systems (EVS)](<https://petronus.eu/blog/synthetic-conscience-the-emergence-of-engineered-vitality-systems-evs/>)

- [The Synthetic Conscience Effect: How ΔE Translates Awareness into Engineering](<https://petronus.eu/blog/the-synthetic-conscience-effect-how-de-translates-awareness-into-engineering/>)

Behavioral Core:

- [Impulse $\rightarrow$ Awareness $\rightarrow$ Coherence: A Unified Logic of Behaviour for Any Adaptive System](<https://petronus.eu/blog/impulse-awareness-coherence-a-unified-logic-of-behaviour-for-any-adaptive/>)

- [Structural Drift as a Fundamental Law of Adaptive Behavior](<https://petronus.eu/blog/structural-drift-as-a-fundamental-law-of-adaptive-behavior/>)

UTAM Series:

- [Unified Theory of Adaptive Meaning — Part I](<https://petronus.eu/blog/unified-theory-of-adaptive-meaning-utam-will-coherence-and-drift-as-the/>)

- Unified Theory of Adaptive Meaning — Part II (this document)

ONTO Σ Series:

- ONTO Σ I–IX, published on [petronus.eu/works/](<https://petronus.eu/works/>)

- [ONTO Σ X — The Pulsating Interior](<https://petronus.eu/blog/onto-x-pulsating-interior/>) (DOI: 10.5281/zenodo.19614567)

- [ONTO Σ VII.1 — Verification as Admissibility Examination](<https://petronus.eu/blog/onto-vii-1-verification-not-causal/>) (DOI: 10.5281/zenodo.19609707)

Engineering & Bridge Works:

- [Structural Pressure: The Missing Primitive](<https://petronus.eu/blog/structural-pressure-the-missing-primitive/>)

- [Transaction vs Structural Admissibility](<https://petronus.eu/blog/transaction-vs-structural-admissibility/>)

- [Coordination Computation Class](<https://petronus.eu/blog/coordination-computation-class/>)

- [Structural Navigation Agent](<https://petronus.eu/blog/structural-navigation-agent/>)

- [Continuity-Bounded Coordination](<https://petronus.eu/blog/continuity-bounded-coordination/>)

- [Why Enforcement Requires Non-Participation](<https://petronus.eu/blog/enforcement-non-participation/>)

- [When Agreement Means Something](<https://petronus.eu/blog/when-agreement-means-something/>)

- [Memory as System Depth](<https://petronus.eu/blog/memory-as-system-depth/>)

- [The Brain Does Not Optimize Truth — It Navigates Admissible Regimes](<https://petronus.eu/blog/the-brain-does-not-optimize-truth-it-navigates-admissible-regimes/>)

- [Why Causality Is Not Enough](<https://petronus.eu/blog/why-causality-is-not-enough/>)

- [Subtle Substitution: On the Drift of Reality in the Age of Algorithmic Mediation](<https://petronus.eu/blog/subtle-substitution-on-the-drift-of-reality-in-the-age-of-algorithmic-mediation/>)

Extremum Series:

- Extremum I–IV, published on [petronus.eu/works/](<https://petronus.eu/works/>)

- [Extremum V — Self-Induced Structural Depletion](<https://petronus.eu/blog/extremum-v-self-induced-depletion/>) (DOI: 10.5281/zenodo.19617127)

- [Extremum VI — Asymmetric Temporal Exhaustion](<https://petronus.eu/blog/extremum-vi-asymmetric-temporal-exhaustion/>)

Experiential:

- Through a Life — Parts I–IV, published on [petronus.eu/works/](<https://petronus.eu/works/>)

Axiomatic Core:

- Navigational Cybernetics 2.5, v2.1 — [DOI: 10.17605/OSF.IO/NHTC5](https://doi.org/10.17605/OSF.IO/NHTC5)

- Full corpus: petronus.eu/works

Appendix — Theorem Navigation Map

This appendix provides direct traceability from each bridge claim in UTAM Part II to the formal apparatus of NC2.5 v2.1 (DOI: 10.17605/OSF.IO/NHTC5). Format: Claim → Section → NC2.5 Reference → Dependencies → Counterexample Form.

Claim in UTAM Part II	Section	NC2.5 v2.1 Reference	Dependencies	Counterexample Form
Admissibility is non-causal	1	Axiom 29 (Non-Causality of Admissibility)	Primitive axiom	System where $\text{Adm}(\cdot)$ emits gradient signal or shapes reward
Admissibility precedes evaluation	1	Axiom 31 (Authorization-before-Evaluation)	Axiom 29	System where evaluation precedes admissibility check and produces equivalent outcomes
Meaning-preserving manifold = E_{adm}	1	Bridge result (this document)	Axiom 29, definition of E_{adm}	System where $M \neq E_{\text{adm}}$: meaning-preserving trajectory that is inadmissible, or admissible trajectory that violates meaning
Structural burden Φ is monotone	2	Axiom 27 (Inevitability of Structural Consumption)	Axioms 6, 7	Transition that decreases Φ
Volume of Will = $\tau = C - \Phi$	3	Axiom 7 (Internal Time Bounds Validity)	Primitive axiom	System with $C = \infty$ (unbounded budget)
Finite structural horizon	3	Theorem 63 (Pressure-Induced Finite Horizon)	Axioms 7, 27, 61	Bounded coupled system with verified $P_{\text{min}} > 0$, observed $\tau(t) > 0$ beyond $t^* = \tau(0)/P_{\text{min}}$ at $U = 0$
Structural pressure at zero action	3	Axiom 61 (Structural Pressure)	Axiom 27, Coupling Criterion	Coupled system with $P = 0$ under verified non-zero coupling
Spin is necessary	4	Theorem 62 (Spin Necessity)	Axiom 7, Lemma 16	Bounded- τ system with purely potential dynamics and non-stagnant identity over $T > 10/P_{\text{min}}$

Gradient collapse on bounded orbits	4	Lemma 16 (Gradient Collapse)	LaSalle invariance principle (external)	Bounded potential system where $\nabla V \neq 0$ on ω -limit set
Helmholtz decomposition of state velocity	4	Applied result	Helmholtz 1858 (external)	Vector field that admits no unique decomposition into gradient + divergence-free
Admissibility boundary non-reconstructible	5	Axiom 51 (Non-Rec onstructible Boundary)	Axiom 29	External observer reconstructing E_{adm} geometry with precision $< \varepsilon$
NR- ε information-theoretic bound	5	Derived from Axiom 51	Fano 1961, Pinsker 1964 (external)	Observation protocol that distinguishes boundary hypotheses below declared KL bound
Meta-revision bounded by Lyapunov descent	6	Axiom 55 (Revision Lyapunov Function)	Axioms 53, 54	Meta-revision sequence that does not converge (oscillates indefinitely)
Structural closure of meta-revision	6	Theorem 55 (Structural Closure)	Axioms 49–55, 58–59	Leakage path from meta-revision into optimization surface
Meta-revision consumes τ	6	Theorem 59 (Double-Budget Closure)	Theorems 55, 58	Meta-revision step with $\Delta\Phi = 0$
Drift is inevitable	2, 4	Axiom 6 (Inevitability of Drift)	Primitive axiom	Adaptive system with zero drift over unbounded horizon
Drift is navigable	8	Axiom 44 (Drift as Navigable Medium)	Axiom 6	Drifting system where no directional choice affects drift trajectory
Identity $\neq$ Performance	0.5	Axiom 13 (Identity $\neq$ Performance)	Primitive axiom	System where sustained performance implies sustained identity
Non-stagnant identity under bounded orbit requires spin	7	Corollary 62.2 (Spin Necessity for Level 3)	Theorem 62, Regime Depth definition	Level 3 system maintaining attractor-exclusion without non-potential component
Class is non-empty	7	$T^1 \times \blacksquare\blacksquare$ witness	Axioms 7, 27, Theorem 62	— (existence proof; no counterexample applicable)
IIC = $W \rightarrow E \rightarrow A$ in time	4.7	Bridge result (this document)	IIC Law + $W \rightarrow E \rightarrow A$ triad	Meaning-preserving system that does not exhibit three-phase cycle

ΔE reduces rate of Φ accumulation	4.5	Architectural claim	Axioms 6, 27	System with ΔE where Φ accumulation rate equals or exceeds no- ΔE baseline
--	-----	---------------------	--------------	---

How to use this map. Each row links a prose claim in the bridge text to its formal anchor. To verify a claim: locate the NC2.5 v2.1 axiom or theorem by number, check its dependency chain, and attempt the counterexample form. If the counterexample is realized within the declared architectural class, the claim is falsified. If the dependency is dropped, the claim weakens to the stated degree.

This map does not add new results. It provides navigation.

Maksim Barziankou (MxBv)

PETRONUS™ | Navigational Cybernetics 2.5

petronus.eu

This work is part of the NC2.5 corpus.

DOI: 10.5281/zenodo.19646174

Axiomatic Core: 10.17605/OSF.IO/NHTC5

License: CC BY-NC-ND 4.0

Maksim Barziankou (MxBv) · PETRONUS · research@petronus.eu
petronus.eu | DOI: 10.5281/zenodo.19646174
Axiomatic Core (NC2.5 v2.1): 10.17605/OSF.IO/NHTC5

CC BY-NC-ND 4.0 · No commercial use, no derivatives.